

Advanced Optimization

Chapter 5 — Conjugate Gradient Methods

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Reading: Nocedal & Wright, *Numerical Optimization*, 2nd ed., Chapter 5

Conjugate gradient methods: two uses

Linear CG (§5.1)

Hestenes and Stiefel, 1950s. Solves $Ax = b$, A symmetric positive definite, without factorizing A . Speed depends on the **eigenvalue distribution** of A ; preconditioning improves it.

Linear CG returns later in the course:

- ▶ Chapter 7: Newton–CG and CG–Steihaug (trust region).
- ▶ Chapter 16: projected CG for equality-constrained QP.

Read: pp. 101–102.

Nonlinear CG (§5.2)

Fletcher and Reeves, 1960s. One of the earliest large-scale methods. No matrix storage, faster than steepest descent.

One problem, two views

For $A \in \mathbb{R}^{n \times n}$ symmetric positive definite, these have the same unique solution x^* :

$$Ax = b \quad (5.1) \quad \Longleftrightarrow \quad \min_x \phi(x) \stackrel{\text{def}}{=} \frac{1}{2}x^T Ax - b^T x. \quad (5.2)$$

The gradient of ϕ is the residual of the linear system:

$$\nabla \phi(x) = Ax - b \stackrel{\text{def}}{=} r(x), \quad r_k = Ax_k - b. \quad (5.3, 5.4)$$

Reading this chapter

“Residual” and “gradient” are the same thing. “Minimize ϕ along p ” is an exact line search on a quadratic.

Conjugate directions

Definition

Nonzero vectors $\{p_0, p_1, \dots, p_l\}$ are **conjugate** with respect to A if

$$p_i^T A p_j = 0 \quad \text{for all } i \neq j. \quad (5.5)$$

A conjugate set is linearly independent (Exercise 5.2), so there are at most n of them.

Conjugate direction method. Given x_0 and conjugate $\{p_0, \dots, p_{n-1}\}$:

$$x_{k+1} = x_k + \alpha_k p_k, \quad \alpha_k = -\frac{r_k^T p_k}{p_k^T A p_k}. \quad (5.6, 5.7)$$

α_k is the exact minimizer of ϕ along $x_k + \alpha p_k$ (see (3.55)).

Theorem 5.1: at most n steps

Theorem 5.1

For any $x_0 \in \mathbb{R}^n$, the sequence $\{x_k\}$ generated by (5.6), (5.7) converges to the solution x^* of (5.1) in at most n steps.

Proof. The p_i span $\mathbb{R}^n$: $x^* - x_0 = \sigma_0 p_0 + \cdots + \sigma_{n-1} p_{n-1}$. Multiply by $p_k^T A$; conjugacy leaves one term:

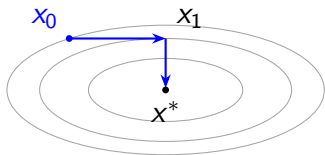
$$\sigma_k = \frac{p_k^T A(x^* - x_0)}{p_k^T A p_k}. \quad (5.8)$$

Since $x_k = x_0 + \alpha_0 p_0 + \cdots + \alpha_{k-1} p_{k-1}$, conjugacy gives $p_k^T A(x_k - x_0) = 0$, so

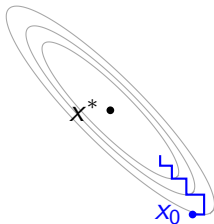
$$p_k^T A(x^* - x_0) = p_k^T A(x^* - x_k) = p_k^T (b - Ax_k) = -p_k^T r_k.$$

Compare with (5.7): $\sigma_k = \alpha_k$. □

The geometry



A diagonal: done in n steps (cf. Figure 5.1)



General A : not finite (cf. Figure 5.2)

Fix: change variables. With $S = [p_0 \ p_1 \ \cdots \ p_{n-1}]$ and $\hat{x} = S^{-1}x$ (5.9), $\hat{\phi}(\hat{x}) = \frac{1}{2}\hat{x}^T(S^TAS)\hat{x} - (S^Tb)^T\hat{x}$, and S^TAS is diagonal. Coordinate search in $\hat{x}$ is the conjugate direction method in x . **Read:** pp. 102–106.

Theorem 5.2: expanding subspace minimization

From (5.4) and (5.6): $r_{k+1} = r_k + \alpha_k A p_k$. (5.10)

Theorem 5.2 (Expanding Subspace Minimization)

For any x_0 , the iterates of the conjugate direction method (5.6), (5.7) satisfy

$$r_k^T p_i = 0, \quad i = 0, 1, \dots, k-1, \quad (5.11)$$

and x_k minimizes ϕ over the set

$$\{x \mid x = x_0 + \text{span}\{p_0, p_1, \dots, p_{k-1}\}\}. \quad (5.12)$$

After k steps, x_k is **optimal on a k -dimensional subspace**: useful even when stopped early.

Proof of Theorem 5.2

Optimality. Let $h(\sigma) = \phi(x_0 + \sigma_0 p_0 + \cdots + \sigma_{k-1} p_{k-1})$, a strictly convex quadratic. Its minimizer satisfies $\partial h / \partial \sigma_i = 0$, that is, $r(\tilde{x})^T p_i = 0$ for $i < k$. So a point minimizes ϕ over (5.12) iff its residual is orthogonal to $p_0, \dots, p_{k-1}$.

Induction on (5.11). $k = 1$: x_1 minimizes ϕ along p_0 , so $r_1^T p_0 = 0$. Assume $r_{k-1}^T p_i = 0$ for $i \leq k - 2$. By (5.10),

$$\begin{aligned} p_{k-1}^T r_k &= p_{k-1}^T r_{k-1} + \alpha_{k-1} p_{k-1}^T A p_{k-1} = 0 && \text{by the choice (5.7) of } \alpha_{k-1}, \\ p_i^T r_k &= p_i^T r_{k-1} + \alpha_{k-1} p_i^T A p_{k-1} = 0, \quad i \leq k - 2 && \text{by hypothesis and conjugacy.} \end{aligned}$$



Generating the directions on the fly

Any conjugate set works: the eigenvectors of A , or Gram–Schmidt in the A -inner product. But eigenvectors cost too much, and Gram–Schmidt must store all previous directions.

CG: use only the steepest descent direction $-r_k$ and the previous direction:

$$p_k = -r_k + \beta_k p_{k-1}. \quad (5.13)$$

Require $p_{k-1}^T A p_k = 0$:

$$\beta_k = \frac{r_k^T A p_{k-1}}{p_{k-1}^T A p_{k-1}}.$$

Start with $p_0 = -r_0$, the steepest descent direction at x_0 .

The surprise

We imposed conjugacy only with p_{k-1} , yet p_k is conjugate to *all* of $p_0, \dots, p_{k-1}$ (Theorem 5.3).

Algorithm 5.1 (CG, preliminary version)

Given x_0 ; $r_0 \leftarrow Ax_0 - b$, $p_0 \leftarrow -r_0$, $k \leftarrow 0$;

while $r_k \neq 0$

$$\alpha_k \leftarrow -\frac{r_k^T p_k}{p_k^T A p_k}; \quad (5.14a)$$

$$x_{k+1} \leftarrow x_k + \alpha_k p_k; \quad (5.14b)$$

$$r_{k+1} \leftarrow Ax_{k+1} - b; \quad (5.14c)$$

$$\beta_{k+1} \leftarrow \frac{r_{k+1}^T A p_k}{p_k^T A p_k}; \quad (5.14d)$$

$$p_{k+1} \leftarrow -r_{k+1} + \beta_{k+1} p_k; \quad (5.14e)$$

$$k \leftarrow k + 1; \quad (5.14f)$$

end while

Useful for proving properties. A cheaper version follows (Algorithm 5.2).

Theorem 5.3: the four properties

Krylov subspace of degree k for r_0 :

$$\mathcal{K}(r_0; k) \stackrel{\text{def}}{=} \text{span}\{r_0, Ar_0, \dots, A^k r_0\}. \quad (5.15)$$

Theorem 5.3

If the k th iterate of CG is not x^* , then

$$r_k^T r_i = 0, \quad i = 0, \dots, k-1, \quad (5.16)$$

$$\text{span}\{r_0, r_1, \dots, r_k\} = \mathcal{K}(r_0; k), \quad (5.17)$$

$$\text{span}\{p_0, p_1, \dots, p_k\} = \mathcal{K}(r_0; k), \quad (5.18)$$

$$p_k^T A p_i = 0, \quad i = 0, \dots, k-1. \quad (5.19)$$

Therefore $\{x_k\}$ converges to x^* in at most n steps.

Proof of Theorem 5.3 (outline)

Induction on k for (5.17)–(5.19).

- ▶ (5.17): $Ap_k \in \text{span}\{Ar_0, \dots, A^{k+1}r_0\}$ (5.20), then (5.10) gives $r_{k+1} \in \mathcal{K}(r_0; k+1)$. Conversely $A^{k+1}r_0 \in \text{span}\{Ap_0, \dots, Ap_k\}$ and $Ap_i = (r_{i+1} - r_i)/\alpha_i$.
- ▶ (5.18): by (5.14e), then (5.17).
- ▶ (5.19): multiply (5.14e) by Ap_i :

$$p_{k+1}^T Ap_i = -r_{k+1}^T Ap_i + \beta_{k+1} p_k^T Ap_i. \quad (5.21)$$

$i = k$: zero by the choice of β_{k+1} . $i < k$: Theorem 5.2 gives $r_{k+1}^T p_j = 0$, $j \leq k$ (5.22), and $Ap_i \in \text{span}\{p_0, \dots, p_{i+1}\}$ (5.23).

(5.16): $r_i \in \text{span}\{p_i, p_{i-1}\}$, so (5.11) gives $r_k^T r_i = 0$.

The proof needs $p_0 = -r_0$. With another p_0 the result fails. **Read:** pp. 107–111.

Remarks on Theorem 5.3

- ▶ Residuals are mutually **orthogonal**; directions are mutually **conjugate**.
“Conjugate gradient” is a misnomer: it is the directions, not the gradients, that are conjugate.
- ▶ CG is a **Krylov subspace method**: every iterate lies in $x_0 + \mathcal{K}(r_0; k)$. This view gives the convergence rate.
- ▶ Only p_{k-1} is needed to compute p_k : little storage and work.

Algorithm 5.2 (CG)

By (5.14e) and (5.11): $r_k^T p_k = -r_k^T r_k$. By (5.10) and (5.16):
 $\alpha_k r_{k+1}^T A p_k = r_{k+1}^T r_{k+1}$. Hence

Given x_0 ; $r_0 \leftarrow Ax_0 - b$, $p_0 \leftarrow -r_0$, $k \leftarrow 0$;

while $r_k \neq 0$

$$\alpha_k \leftarrow \frac{r_k^T r_k}{p_k^T A p_k}; \quad x_{k+1} \leftarrow x_k + \alpha_k p_k; \quad r_{k+1} \leftarrow r_k + \alpha_k A p_k; \quad (5.24a-c)$$

$$\beta_{k+1} \leftarrow \frac{r_{k+1}^T r_{k+1}}{r_k^T r_k}; \quad p_{k+1} \leftarrow -r_{k+1} + \beta_{k+1} p_k; \quad k \leftarrow k + 1; \quad (5.24d-f)$$

end while

Read: pp. 111–112.

Algorithm 5.2: cost and use

- ▶ Per iteration: one product Ap_k , two inner products $p_k^T(Ap_k)$ and $r_{k+1}^T r_{k+1}$, three vector updates.
- ▶ Only x , r , p from the last iteration are kept: overwrite in place.
- ▶ A is never modified or factored: **no fill-in**. A can be given as a function $p \mapsto Ap$.
- ▶ For small n , use Gaussian elimination or another factorization: less sensitive to rounding. CG is recommended only for large problems.
- ▶ CG sometimes approaches the solution in far fewer than n steps. Why? Next: the rate of convergence.

The polynomial view

From (5.24b) and (5.18), for some polynomial P_k^* of degree k :

$$x_{k+1} = x_0 + \gamma_0 r_0 + \gamma_1 A r_0 + \cdots + \gamma_k A^k r_0 = x_0 + P_k^*(A) r_0. \quad (5.25, 5.26)$$

With $\|z\|_A^2 = z^T A z$ (5.27), the same norm as in Chapter 3,

$$\frac{1}{2} \|x - x^*\|_A^2 = \phi(x) - \phi(x^*). \quad (5.28)$$

By Theorem 5.2, x_{k+1} minimizes ϕ over $x_0 + \mathcal{K}(r_0; k)$. Hence

$$\|x_{k+1} - x^*\|_A = \min_{P_k} \|x_0 + P_k(A) r_0 - x^*\|_A. \quad (5.29)$$

CG is optimal among all methods whose steps stay in the Krylov subspace.

Expanding in eigenvectors

$r_0 = A(x_0 - x^*)$, so $x_{k+1} - x^* = [I + P_k^*(A)A](x_0 - x^*)$ (5.30). With eigenpairs $0 < \lambda_1 \leq \dots \leq \lambda_n$, v_i , and $x_0 - x^* = \sum_i \xi_i v_i$ (5.31):

$$\|x_{k+1} - x^*\|_A^2 = \sum_{i=1}^n \lambda_i [1 + \lambda_i P_k^*(\lambda_i)]^2 \xi_i^2. \quad (5.32)$$

P_k^* is optimal, so

$$\|x_{k+1} - x^*\|_A^2 \leq \min_{P_k} \max_{1 \leq i \leq n} [1 + \lambda_i P_k(\lambda_i)]^2 \|x_0 - x^*\|_A^2. \quad (5.33)$$

The rate in one sentence

CG is as fast as a degree- $(k+1)$ polynomial Q with $Q(0) = 1$ can be made small on the eigenvalues of A . **Where the eigenvalues are** matters, not n .

Theorem 5.4: distinct eigenvalues

Theorem 5.4

If A has only r distinct eigenvalues, then CG terminates at the solution in at most r iterations.

Proof. Let the distinct eigenvalues be $\tau_1 < \dots < \tau_r$ and

$$Q_r(\lambda) = \frac{(-1)^r}{\tau_1 \tau_2 \cdots \tau_r} (\lambda - \tau_1)(\lambda - \tau_2) \cdots (\lambda - \tau_r).$$

Then $Q_r(\lambda_i) = 0$ for all i and $Q_r(0) = 1$, so $\bar{P}_{r-1}(\lambda) = (Q_r(\lambda) - 1)/\lambda$ is a polynomial of degree $r - 1$. In (5.34) with $k = r - 1$:

$$0 \leq \min_{P_{r-1}} \max_i [1 + \lambda_i P_{r-1}(\lambda_i)]^2 \leq \max_i Q_r^2(\lambda_i) = 0.$$

By (5.33), $x_r = x^*$.



Theorem 5.5: clusters

Theorem 5.5

If A has eigenvalues $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$, then

$$\|x_{k+1} - x^*\|_A^2 \leq \left(\frac{\lambda_{n-k} - \lambda_1}{\lambda_{n-k} + \lambda_1} \right)^2 \|x_0 - x^*\|_A^2. \quad (5.35)$$

Idea: Q_{k+1} with roots at the k largest eigenvalues and at the midpoint of $[\lambda_1, \lambda_{n-k}]$.

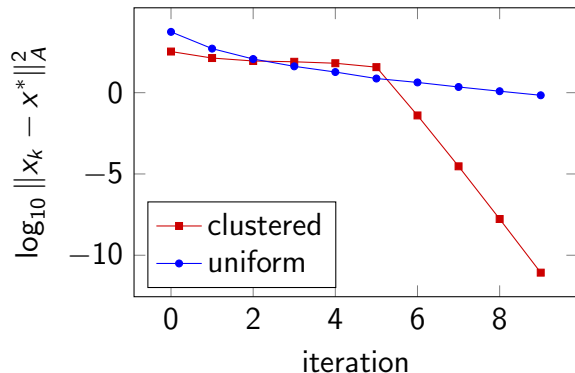


cf. Figure 5.3

More generally: r clusters $\Rightarrow$ about r iterations.

m large eigenvalues, the rest in a cluster of width ϵ near 1: after $m + 1$ steps,
 $\|x_{m+1} - x^*\|_A \approx \epsilon \|x_0 - x^*\|_A.$

Numerical example: clustered eigenvalues



$n = 100$. (Not in the book; cf. Figure 5.4.)

- ▶ Clustered: 5 eigenvalues $20, \dots, 100$, the other 95 in $[0.95, 1.05]$. Sharp drop at iteration $6 = m + 1$, as Theorem 5.5 predicts.
- ▶ Uniform: eigenvalues spread over $[1, 100]$. Slow, steady decrease.

The condition-number bound

If only λ_1 and λ_n are known, with $\kappa(A) = \lambda_n/\lambda_1$:

$$\|x_k - x^*\|_A \leq 2 \left(\frac{\sqrt{\kappa(A)} - 1}{\sqrt{\kappa(A)} + 1} \right)^k \|x_0 - x^*\|_A. \quad (5.36)$$

Steepest descent (3.29) has the same form with κ in place of $\sqrt{\kappa}$.

κ	SD: iterations to reduce error by 10^6	CG
10^2	≈ 690	≈ 70
10^4	$\approx 69,000$	≈ 690
10^6	$\approx 6.9 \times 10^6$	$\approx 6,900$

(5.36) is often a large overestimate; the eigenvalue distribution (Theorem 5.5) tells more. **Read: pp. 112–118.**

Numerical example: CG in floating point

Exercise 5.1: Hilbert matrix $A_{ij} = 1/(i + j - 1)$, $b = (1, \dots, 1)^T$, $x_0 = 0$, stop when $\|r_k\| \leq 10^{-6}\|b\|$.

n	$\kappa(A)$	CG iterations
5	4.8×10^5	6
8	1.5×10^{10}	19
12	1.8×10^{16}	37
20	1.0×10^{19}	68

More than n iterations: in floating point the directions lose conjugacy, and Theorem 5.1 no longer applies. CG then behaves as an iterative method, and the rate theory still describes it. (Not in the book.)

Preconditioning

Change variables with a nonsingular C :

$$\hat{x} = Cx, \quad \hat{\phi}(\hat{x}) = \frac{1}{2}\hat{x}^T(C^{-T}AC^{-1})\hat{x} - (C^{-T}b)^T\hat{x}. \quad (5.37, 5.38)$$

CG on $\hat{\phi}$ converges according to the eigenvalues of $C^{-T}AC^{-1}$, not of A .

Goal

Choose C so that $C^{-T}AC^{-1}$ has a small condition number, or better, clustered eigenvalues.

The transformation is never carried out. Run CG in $\hat{x}$, translate back to x : only $M = C^T C$ (symmetric positive definite) appears, and only through solves $My = r$.

Algorithm 5.3 (Preconditioned CG)

Given x_0 , preconditioner M ; $r_0 \leftarrow Ax_0 - b$; solve $My_0 = r_0$; $p_0 \leftarrow -y_0$; $k \leftarrow 0$;

while $r_k \neq 0$

$$\alpha_k \leftarrow \frac{r_k^T y_k}{p_k^T A p_k}; \quad x_{k+1} \leftarrow x_k + \alpha_k p_k; \quad r_{k+1} \leftarrow r_k + \alpha_k A p_k; \quad (5.39a-c)$$

$$\text{solve } My_{k+1} = r_{k+1}; \quad (5.39d)$$

$$\beta_{k+1} \leftarrow \frac{r_{k+1}^T y_{k+1}}{r_k^T y_k}; \quad p_{k+1} \leftarrow -y_{k+1} + \beta_{k+1} p_k; \quad k \leftarrow k + 1; \quad (5.39e-g)$$

end while

- ▶ $M = I$ gives Algorithm 5.2.
- ▶ Orthogonality (5.16) becomes $r_i^T M^{-1} r_j = 0$ for $i \neq j$. (5.40)
- ▶ Extra cost: one solve with M per iteration. It must be cheap.

Practical preconditioners

No single choice is best. Trade-off: effectiveness of M , cost of forming and storing M , cost of solving $My = r$.

- ▶ **Problem-specific:** for a PDE, $My = r$ can be a coarser discretization of the same problem. Knowing where A comes from is the key.
- ▶ **General purpose:** SSOR, banded, and **incomplete Cholesky** (probably the most effective). Compute $\tilde{L}$ no denser than the lower triangle of A , $A \approx \tilde{L}\tilde{L}^T$, $C = \tilde{L}^T$:

$$C^{-T}AC^{-1} = \tilde{L}^{-1}A\tilde{L}^{-T} \approx I.$$

$My = r$ is two triangular solves, about the cost of one Ap .

- ▶ **Pitfalls:** $\tilde{L}\tilde{L}^T$ may not be positive definite (increase the diagonal); breakdown can occur (allow more fill, at higher cost).

Read: pp. 118–120.

Numerical example: incomplete Cholesky

2-D Poisson equation, 5-point stencil on a 60×60 grid ($n = 3600$). Stop when $\|r_k\| \leq 10^{-8} \|b\|$. (Not in the book.)

	condition number	iterations
CG, $M = I$	$\kappa(A) = 1507$	112
PCG, $M = \tilde{L}\tilde{L}^T$, IC(0)	$\kappa(\tilde{L}^{-1}A\tilde{L}^{-T}) = 134$	49

- ▶ IC(0): $\tilde{L}$ has the sparsity pattern of the lower triangle of A ; no fill-in.
- ▶ κ drops by a factor 11, so $\sqrt{\kappa}$ by about 3.4. The iteration count drops by 2.3.

Algorithm 5.4 (Fletcher–Reeves)

Two changes to Algorithm 5.2: a line search on f replaces (5.24a); ∇f replaces r .

Given x_0 ; evaluate $f_0 = f(x_0)$, ∇f_0 ; $p_0 \leftarrow -\nabla f_0$; $k \leftarrow 0$;

while $\nabla f_k \neq 0$

 compute α_k by a line search; $x_{k+1} = x_k + \alpha_k p_k$; evaluate ∇f_{k+1} ;

$$\beta_{k+1}^{\text{FR}} \leftarrow \frac{\nabla f_{k+1}^T \nabla f_{k+1}}{\nabla f_k^T \nabla f_k}; \quad (5.41a)$$

$$p_{k+1} \leftarrow -\nabla f_{k+1} + \beta_{k+1}^{\text{FR}} p_k; \quad (5.41b)$$

$$k \leftarrow k + 1; \quad (5.41c)$$

end while

- ▶ Strongly convex quadratic + exact line search: this is Algorithm 5.2.
- ▶ Each iteration needs only f and ∇f . No matrices, a few vectors.

Is p_k a descent direction?

Take the inner product of (5.41b) with ∇f_k :

$$\nabla f_k^T p_k = -\|\nabla f_k\|^2 + \beta_k^{\text{FR}} \nabla f_k^T p_{k-1}. \quad (5.42)$$

Exact line search: $\nabla f_k^T p_{k-1} = 0$, so descent. Inexact: p_k may be an **ascent direction**.

Strong Wolfe conditions with $c_2 < \frac{1}{2}$

$$f(x_k + \alpha_k p_k) \leq f(x_k) + c_1 \alpha_k \nabla f_k^T p_k, \quad (5.43a)$$

$$|\nabla f(x_k + \alpha_k p_k)^T p_k| \leq -c_2 \nabla f_k^T p_k, \quad (5.43b)$$

with $0 < c_1 < c_2 < \frac{1}{2}$, not $c_2 < 1$ as in (3.7). Then every p_k is a descent direction (Lemma 5.6).

Read: pp. 121–122.

Polak–Ribière and variants

$$\beta_{k+1}^{\text{PR}} = \frac{\nabla f_{k+1}^T (\nabla f_{k+1} - \nabla f_k)}{\|\nabla f_k\|^2}, \quad (5.44)$$

$$\beta_{k+1}^+ = \max\{\beta_{k+1}^{\text{PR}}, 0\} \quad (\text{PR+}), \quad (5.45)$$

$$\beta_{k+1}^{\text{HS}} = \frac{\nabla f_{k+1}^T (\nabla f_{k+1} - \nabla f_k)}{(\nabla f_{k+1} - \nabla f_k)^T p_k}. \quad (5.46)$$

- ▶ All equal β^{FR} on a strongly convex quadratic with exact line search, since then $\nabla f_{k+1}^T \nabla f_k = 0$ (5.16).
- ▶ PR: more robust and efficient in practice. But strong Wolfe does *not* guarantee descent for PR.
- ▶ HS: $p_{k+1}^T \bar{G}_k p_k = 0$ with the average Hessian $\bar{G}_k = \int_0^1 \nabla^2 f(x_k + \tau \alpha_k p_k) d\tau$, since $\nabla f_{k+1} = \nabla f_k + \alpha_k \bar{G}_k p_k$.

More choices of β

Global convergence holds for any β_k with

$$|\beta_k| \leq \beta_k^{\text{FR}}, \quad k \geq 2. \quad (5.47)$$

$$\mathbf{FR-PR}: \text{PR clipped to } [-\beta_k^{\text{FR}}, \beta_k^{\text{FR}}] \quad (5.48)$$

Two recent choices, with $\hat{y}_k = \nabla f_{k+1} - \nabla f_k$:

$$\beta_{k+1} = \frac{\|\nabla f_{k+1}\|^2}{\hat{y}_k^T p_k} \quad (\text{Dai-Yuan}), \quad (5.49)$$

$$\beta_{k+1} = \left(\hat{y}_k - 2p_k \frac{\|\hat{y}_k\|^2}{\hat{y}_k^T p_k} \right)^T \frac{\nabla f_{k+1}}{\hat{y}_k^T p_k} \quad (\text{Hager-Zhang}). \quad (5.50)$$

Both give descent directions under the (ordinary) Wolfe conditions, and are competitive with PR. **Read: pp. 122–124.**

Quadratic termination and restarts

- ▶ Line searches use quadratic or cubic interpolation (Chapter 3): on a quadratic, α_k is exact and the method is Algorithm 5.2.
- ▶ **Restart every n steps** ($\beta_k = 0$): n -step quadratic convergence,

$$\|x_{k+n} - x^*\| = O(\|x_k - x^*\|^2). \quad (5.51)$$

Near x^* , f is nearly quadratic; after a restart there, the method is nearly linear CG, which needs $p_0 = -\nabla f$.

- ▶ But CG is for large n , and a solution is often found in fewer than n steps. Restart instead when consecutive gradients are far from orthogonal:

$$\frac{|\nabla f_k^T \nabla f_{k-1}|}{\|\nabla f_k\|^2} \geq \nu, \quad \nu \approx 0.1. \quad (5.52)$$

- ▶ PR+ (5.45) is also a restart: $p_{k+1} = -\nabla f_{k+1}$ whenever $\beta^{\text{PR}} < 0$. This is rare.

Read: pp. 124–125.

Lemma 5.6: FR gives descent directions

Lemma 5.6

Suppose Algorithm 5.4 uses step lengths that satisfy the strong Wolfe conditions (5.43) with $0 < c_2 < \frac{1}{2}$. Then every p_k is a descent direction, and

$$-\frac{1}{1-c_2} \leq \frac{\nabla f_k^T p_k}{\|\nabla f_k\|^2} \leq \frac{2c_2-1}{1-c_2}, \quad k = 0, 1, \dots \quad (5.53)$$

Proof idea. $t(\xi) = (2\xi - 1)/(1 - \xi)$ increases on $[0, \frac{1}{2}]$ from -1 to 0 , so the right side is in $(-1, 0)$ (5.54). Induction: by (5.41),

$$\frac{\nabla f_{k+1}^T p_{k+1}}{\|\nabla f_{k+1}\|^2} = -1 + \frac{\nabla f_{k+1}^T p_k}{\|\nabla f_k\|^2}, \quad (5.55)$$

and (5.43b) bounds the last term by $c_2 |\nabla f_k^T p_k| / \|\nabla f_k\|^2$. Apply (5.53) for k . $\square$

Why FR can stall

With $\cos \theta_k = -\nabla f_k^T p_k / (\|\nabla f_k\| \|p_k\|)$ (5.56), Lemma 5.6 gives

$$\frac{1 - 2c_2}{1 - c_2} \frac{\|\nabla f_k\|}{\|p_k\|} \leq \cos \theta_k \leq \frac{1}{1 - c_2} \frac{\|\nabla f_k\|}{\|p_k\|}. \quad (5.57)$$

So $\cos \theta_k \approx 0$ iff $\|\nabla f_k\| \ll \|p_k\|$: a bad direction, likely a tiny step, and $\nabla f_{k+1} \approx \nabla f_k$.

FR

$\beta_{k+1}^{\text{FR}} \approx 1$ (5.58), so $p_{k+1} \approx p_k$. The bad direction is kept: a long run of useless steps.

PR (also PR+, HS, FR-PR)

$\beta_{k+1}^{\text{PR}} \approx 0$, so $p_{k+1} \approx -\nabla f_{k+1}$. An automatic restart.

Example in [123], $n = 100$: FR needs thousands of iterations, PR needs 37.

Never run FR without restarts. Read: pp. 125–127.

Global convergence with restarts

Assumptions 5.1

- (i) The level set $\mathcal{L} = \{x \mid f(x) \leq f(x_0)\}$ is bounded.
- (ii) f is Lipschitz continuously differentiable in an open neighborhood $\mathcal{N}$ of $\mathcal{L}$.

Then $\|\nabla f(x)\| \leq \bar{\gamma}$ on $\mathcal{L}$ (5.59), and Zoutendijk (Theorem 3.2) gives

$$\sum_{k=0}^{\infty} \cos^2 \theta_k \|\nabla f_k\|^2 < \infty. \quad (5.60)$$

At restart iterations $k_1, k_2, \dots$, $\cos \theta_k = 1$, so $\sum_j \|\nabla f_{k_j}\|^2 < \infty$ (5.61), and

$$\liminf_{k \rightarrow \infty} \|\nabla f_k\| = 0 \quad (5.62)$$

for every method in this chapter, restarted at least every $\bar{n}$ steps. The interesting case: **no restarts**.

Theorem 5.7: Fletcher–Reeves without restarts

Theorem 5.7 (Al-Baali)

Under Assumptions 5.1, Algorithm 5.4 with strong Wolfe steps, $0 < c_1 < c_2 < \frac{1}{2}$, satisfies

$$\liminf_{k \rightarrow \infty} \|\nabla f_k\| = 0. \quad (5.63)$$

Proof idea. Suppose $\|\nabla f_k\| \geq \gamma > 0$ (5.64).

► (5.57) in (5.60): $\sum_k \|\nabla f_k\|^4 / \|p_k\|^2 < \infty$, so $\sum_k 1/\|p_k\|^2 < \infty$. (5.65), (5.70)

► From (5.41b), (5.43b), (5.53), with $c_3 = (1 + c_2)/(1 - c_2)$:
 $\|p_k\|^2 \leq c_3 \|\nabla f_k\|^4 \sum_{j \leq k} \|\nabla f_j\|^{-2} \leq c_3 \bar{\gamma}^4 k / \gamma^2$. (5.67), (5.68)

► Then $\sum_k 1/\|p_k\|^2 \geq \gamma_4 \sum_k 1/k = \infty$. Contradiction. $\square$

Extends to any $|\beta_k| \leq \beta_k^{\text{FR}}$, hence to FR–PR.

Theorem 5.8: Polak–Ribière can cycle

Theorem 5.8 (Powell)

Consider the Polak–Ribière method (5.44) with an ideal line search (the first positive stationary point of $f(x_k + \alpha p_k)$). There exist a twice continuously differentiable $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ and a starting point x_0 such that $\{\nabla f_k\}$ is bounded away from zero.

- ▶ The method that works best in practice has the weakest theory.
- ▶ The proof needs consecutive directions that become almost opposite; with ideal line searches, this happens only if $\beta_k < 0$. Hence **PR+**: reset β_k to 0 when negative.
- ▶ PR+ with a slightly modified Wolfe line search: descent and global convergence.
- ▶ (5.49) and (5.50): global convergence with the standard Wolfe conditions.

Read: pp. 127–131.

Numerical performance

Strong Wolfe, $c_1 = 10^{-4}$, $c_2 = 0.1$, no restarts; stop at $\|\nabla f_k\|_\infty < 10^{-5}(1 + |f_k|)$.
Entries: iterations / function–gradient evaluations.

	Problem, n	FR	PR	PR+
Table 5.1	GENROS, 500	*	1068/2151	1067/2149
	XPOWSING, 1000	533/1102	212/473	97/229
	TRIDIA1, 1000	264/531	262/527	262/527
Our run	ext. Rosenbrock, 1000	68/222	24/105	21/99
	TRIDIA, 1000	669/887	961/1628	961/1628

*: no convergence in 10,000 iterations. “Our run”: SciPy’s strong Wolfe search; not in the book. PR is usually better, but not always. Book’s advice: PR, PR+, FR–PR, or (5.49), (5.50).

Chapter 5 in one table

	Linear CG (§5.1)	Nonlinear CG (§5.2)
Problem	$Ax = b, A \succ 0$	$\min f$
Direction	$p_k = -r_k + \beta_k p_{k-1}$	$p_k = -\nabla f_k + \beta_k p_{k-1}$
Step	exact, $r_k^T r_k / p_k^T A p_k$	strong Wolfe, $c_2 < \frac{1}{2}$
β_k	$r_k^T r_k / r_{k-1}^T r_{k-1}$	FR, PR, PR+, HS, ...
Key property	conjugacy, Krylov optimality	descent (Lemma 5.6)
Termination	$\leq n$; $\leq r$ distinct eigenvalues	$\liminf \ \nabla f_k\ = 0$ (FR)
Rate	$\sqrt{\kappa}$; clusters	n -step quadratic with restarts
Cost	one Ap per iteration	$f, \nabla f$, line search
Accelerator	preconditioning	restarts, choice of β

Three results to carry forward

1. **Conjugacy decouples the problem.** In the basis $\{p_i\}$, an n -dimensional quadratic is n one-dimensional problems (Theorems 5.1, 5.2).
2. **Eigenvalues, not n , set the speed.** r distinct eigenvalues or clusters: about r iterations. Preconditioning changes the distribution.
3. **CG is optimal on the Krylov subspace (5.29).** This is why it can be stopped early, as in Newton–CG and CG–Steihaug (Chapter 7).

For nonlinear CG: strong Wolfe with $c_2 < \frac{1}{2}$, and PR-type β or restarts.

Homework (problems 5–6 are not in the book)

1. Exercise 5.2: a conjugate set is linearly independent.
2. Exercise 5.6: show that (5.24d) is equivalent to (5.14d).
3. Exercise 5.9: derive Algorithm 5.3 from Algorithm 5.2 in the variables $\hat{x}$.
4. Exercise 5.11: PR (5.44) and HS (5.46) reduce to FR on a quadratic with exact line searches.
5. Build $A \in \mathbb{R}^{100 \times 100}$ with exactly 5 distinct eigenvalues. Run Algorithm 5.2 and confirm Theorem 5.4. Then spread each eigenvalue into a cluster of width 10^{-2} and repeat.
6. Implement FR and PR+ with a strong Wolfe line search ($c_2 = 0.1$) on extended Rosenbrock, $n = 100$. Plot $\cos \theta_k$ and β_k .